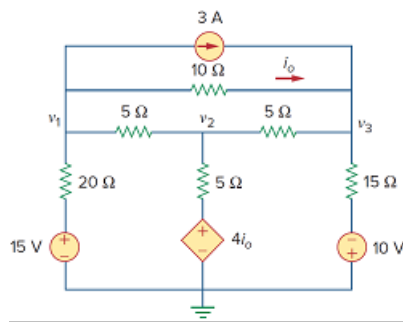


Problem

Calculate the node voltages v_1 , v_2 , and v_3 in the circuit of Fig. 3.75.

Figure 3.75:



Step-by-step solution

Step 1 of 3

By applying KCL at node 1 we can get

$$\frac{v_1 - 15}{20} + \frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{10} + 3 = 0$$

$$v_1 \left[\frac{1}{20} + \frac{1}{5} + \frac{1}{10} \right] - v_2 \left[\frac{1}{5} \right] - v_3 \left[\frac{1}{10} \right] = -3 + \frac{15}{20}$$

$$v_1 [1 + 4 + 2] - v_2 [4] - v_3 [2] = -60 + 15$$

$$7v_1 - 4v_2 - 2v_3 = -45 \dots\dots\dots (1)$$

By applying KCL at node 2 we can get

$$\frac{v_2 - v_1}{5} + \frac{v_2 - 4i_o}{5} + \frac{v_2 - v_3}{5} = 0$$

$$-v_1 \left[\frac{1}{5} \right] + v_2 \left[\frac{1}{5} + \frac{1}{5} + \frac{1}{5} \right] - v_3 \left[\frac{1}{5} \right] = \frac{4i_o}{5}$$

$$-v_1 + 3v_2 - v_3 = 4i_o \dots\dots\dots (2)$$

Step 2 of 3

By applying KCL at node 3 we can write

$$\begin{aligned}\frac{v_3 - v_1}{10} + \frac{v_3 - v_2}{5} + \frac{v_3 + 10}{15} - 3 &= 0 \\ -v_1 \left[\frac{1}{10} \right] - v_2 \left[\frac{1}{5} \right] + v_3 \left[\frac{1}{10} + \frac{1}{5} + \frac{1}{15} \right] &= -\frac{10}{15} + 3 \\ -3v_1 - 6v_2 + v_3 [3 + 6 + 2] &= -20 + 90 \\ -3v_1 - 6v_2 + 11v_3 &= 70 \quad \dots\dots\dots (3)\end{aligned}$$

We can also write

$$i_o = \frac{v_1 - v_3}{10}$$

Now equation (2)

$$\begin{aligned}-v_1 + 3v_2 - v_3 &= 4 \left[\frac{v_1 - v_3}{10} \right] \\ -10v_1 + 30v_2 - 10v_3 &= 4v_1 - 4v_3 \\ 14v_1 - 30v_2 + 6v_3 &= 0 \\ 7v_1 - 15v_2 + 3v_3 &= 0 \quad \dots\dots\dots (4)\end{aligned}$$

Step 3 of 3

We can write equations (1), (3) and (4) in matrix form as below:

$$\begin{bmatrix} 7 & -4 & -2 \\ -3 & -6 & 11 \\ 7 & -15 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -45 \\ 70 \\ 0 \end{bmatrix}$$

We can represent this as

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 7 & -4 & -2 \\ -3 & -6 & 11 \\ 7 & -15 & 3 \end{bmatrix}^{-1} \begin{bmatrix} -45 \\ 70 \\ 0 \end{bmatrix}$$
$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -7.1918 \\ -2.7789 \\ 2.8865 \end{bmatrix}$$

Therefore

$$v_1 = -7.1918$$

$$v_2 = -2.7789$$

$$v_3 = 2.8865$$